

ANTIGRAVITY

Technology Solutions

PROJECT ALLOCATION DOCUMENT

Aynix AI – Student Portal Web Application

Version 1.0 | 20-Day Sprint | April 2025

PROJECT Aynix AI Student Portal	DURATION 20 Working Days	TYPE Full-Stack Web App
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1. Project Overview

Aynix AI is a comprehensive Student Portal web application designed for educational institutes. It serves as a centralized digital hub connecting students, instructors, and administrators. The portal integrates modern AI-powered features, real-time attendance tracking (both online via Google Meet and offline via QR codes), YouTube-based class delivery with access-controlled video playback, and a full-featured admin panel with CRUD operations across all modules.

Project Goal

Build a production-ready, AI-augmented student portal for educational institutes that handles the complete student lifecycle: enrollment, class delivery, attendance, assignments, assessments, communication, and analytics — all within a secure, role-based platform.

1.1 Stakeholders & Roles

Role	Key Responsibilities	Access Level
Super Admin	Full system control, user management, institute configuration	Full Access
Admin	Student/faculty management, class creation, video access control	High Access
Instructor / Faculty	Class delivery, attendance marking, assignments, grading	Medium Access
Student	View classes, attend sessions, submit assignments, view grades	Student Access
Parent (optional)	Monitor attendance, grades, notifications for their ward	Read-Only

1.2 Technology Stack

Frontend

- React.js (with TypeScript)
- Tailwind CSS + shadcn/ui
- React Router v6
- Zustand (state management)
- Socket.io Client (real-time)

Backend

- Node.js + Express.js
- Python/Flask (AI microservices)
- Socket.io (WebSockets)
- REST API + JWT Auth
- Google OAuth 2.0

Database & Storage

- PostgreSQL (primary DB)
- Redis (sessions/cache)
- AWS S3 / Cloudinary (files)

Integrations

- Google Meet API
- YouTube Data API v3
- Google Calendar API
- Gmail API (email delivery)
- OpenAI / Claude API (AI features)
- QR Code libraries (qrcode.js)

2. Attendance Management Module

The attendance module supports two distinct modes of tracking: Online Attendance via Google Meet integration, and Offline Attendance via QR code scanning. Both modes feed into a unified attendance dashboard with real-time analytics, export functionality, and edit/delete capabilities for administrators.

2.1 Online Attendance – Google Meet Integration

How It Works: Google Meet Attendance Tracking

When an instructor starts a class via Google Meet through the portal, the system automatically:

1. Admin/Instructor creates a scheduled class session in the portal (subject, batch, date/time).
2. The portal generates a Google Meet link via the Google Calendar + Meet API and attaches it to the session.
3. Students receive the Meet link on their dashboard (only accessible when logged in with the authorized email).
4. When the Meet session is active, the Google Meet Attendance API tracks participant join/leave timestamps.
5. After the session ends, the portal fetches attendance data from Google Meet API and auto-populates the attendance record.
6. The system computes each student's attendance duration vs. minimum required duration (e.g., 75% of class time).
7. Instructor reviews the auto-generated report, can manually edit/override individual entries, then finalizes.

Google Meet API Integration Details

The portal uses the Google Admin SDK Reports API (admin.reports.v1) to pull Meet attendance data. Specifically, it uses the 'meet' application events to fetch calendar_event_id, organizer_email, participant_email, call_duration_seconds, and meeting_code. OAuth 2.0 service account credentials with domain-wide delegation are required for G-Suite/Workspace accounts.

2.1.1 Online Attendance – Feature Breakdown

Feature	Description	Who Can Use
Auto-Attendance Pull	Automatically fetch attendance from Google Meet after session ends	System (Auto)
Duration Threshold	Mark present/absent based on minimum time attended (configurable %)	Admin
Manual Override	Instructor can edit individual attendance status after auto-pull	Instructor, Admin
Late Join Flag	Flag students who joined after X minutes of class start	Auto / Instructor

Feature	Description	Who Can Use
Attendance Report	Per-session and cumulative report per student/batch	Instructor, Admin
Delete Session Record	Admin can delete attendance record for a session	Admin
Export	Export to Excel/PDF per batch, subject, or date range	Admin, Instructor

2.2 Offline Attendance – QR Code System

How It Works: QR Code Attendance Tracking

For physical/in-person classes, the portal provides a QR-code-based attendance system that is secure, time-bound, and location-aware:

8. Instructor opens the “Start Offline Class” session on their portal dashboard.
9. The system generates a unique, time-limited QR code (valid for 5–10 minutes, refreshing automatically).
10. The QR code is displayed on the classroom projector/screen.
11. Students open the Aynix AI Student Portal on their mobile browser (while logged in), navigate to Attendance, and scan the QR code using their device camera.
12. The backend verifies: (a) QR token validity, (b) student is enrolled in that class/batch, (c) scan is within the valid time window, (d) optionally validates GPS coordinates.
13. Attendance is recorded in real-time. Instructor sees a live dashboard of who has marked attendance.
14. After the window closes, the QR code expires. Admin/instructor can manually add/remove entries.

QR Code Security

Each QR code encodes a signed JWT token containing: session_id, class_id, batch_id, issued_at, expires_at (5–10 min), and a server-side secret. The token is verified server-side. Each token can only be used once per student (replay attack prevention). Optional: device fingerprinting + GPS radius check (configurable per institute).

2.2.1 Offline Attendance – Feature Breakdown

Feature	Description	Who Can Use
QR Generation	Generate unique, rotating, time-bound QR codes per session	Instructor
QR Display Mode	Fullscreen QR display for projector use	Instructor
Student QR Scan	Students scan via mobile browser (no app install needed)	Student

Feature	Description	Who Can Use
Live Dashboard	Real-time view of who scanned, timestamps, pending count	Instructor
Manual Entry	Instructor can manually mark attendance for absent/present	Instructor, Admin
Edit Attendance	Edit any record post-session with reason logging	Admin
Delete Record	Delete attendance record for a session or individual entry	Admin
GPS Validation	Optional: verify student is within campus radius when scanning	Admin (config)
Proxy Prevention	One scan per student per session; duplicate scan blocked	System (Auto)
Cumulative Report	Attendance % per student per subject over a date range	Admin, Instructor

3. Classes & YouTube Video Module

The Classes module enables instructors to deliver course content via YouTube-hosted videos. Access to each video is tightly controlled: Admin grants access per student (by Gmail), and the portal ensures only the authorized student (when logged in with that exact Gmail account) can watch the class video. This provides a private, Institute-controlled streaming experience without making videos public.

3.1 YouTube Video Access Control Flow

How YouTube-Based Class Access Works

15. Instructor records or uploads a class video to YouTube (set to “Unlisted” — not public, not searchable).
16. Instructor submits the YouTube video URL/ID to the portal when creating a class session.
17. Admin reviews and assigns the video to a specific class/subject/batch.
18. Admin grants individual student access by adding the student’s registered Gmail to the access list for that video.
19. An automated email is sent to the student’s Gmail (via Gmail API) containing the portal deep-link to the class, NOT the YouTube URL directly.
20. Student logs into Aynix AI portal with their Gmail account (Google OAuth).
21. The portal checks: logged-in Gmail === authorized Gmail for that video. If matched, the YouTube video is embedded and playable inside the portal.
22. If the student is NOT on the access list or is logged in with a different Gmail, the video is blocked with a “Contact Admin for Access” message.
23. All video views are logged (student, video, watch duration, timestamp) for analytics.

Why Unlisted YouTube + Portal Embedding?

Unlisted YouTube videos are accessible only via direct link — they don’t appear in search or the channel. By embedding them inside the portal (behind Gmail-verified auth), the institute gets: (1) Free video hosting by YouTube, (2) No direct YouTube link exposure to unauthorized users, (3) Full control over who can access which class, (4) Watch analytics inside the portal, (5) Seamless mobile+desktop experience.

3.2 Class Module – Feature Breakdown

Feature	Description	Who Can Use
Create Class	Add class with title, subject, batch, instructor, YouTube URL, description, thumbnail	Admin, Instructor
Edit Class	Edit any class detail including swapping the YouTube video or updating access list	Admin
Delete Class	Soft-delete a class (moves to archive, students lose access)	Admin

Feature	Description	Who Can Use
Access Grant by Gmail	Admin adds student Gmail to approved access list for specific video	Admin
Revoke Access	Remove a student's access to a specific class video	Admin
Bulk Access Grant	Grant/revoke access to an entire batch at once	Admin
Email Notification	Auto-send access email to student's Gmail with portal deep-link	System (Auto)
Gmail-Locked Playback	Video only plays if logged-in Gmail matches authorized Gmail	System (Auto)
Watch Analytics	Track per-student watch duration, completion %, rewatch count	Admin, Instructor
Class Schedule	Attach class to a date/time slot visible on student calendar	Instructor
Class Notes Upload	Attach PDF/PPT notes to a class for student download	Instructor
Student Class Feed	Students see only their accessible classes on dashboard	Student
Playlist / Batch View	Group classes by subject/batch as a course playlist	Student, Instructor

3.3 Student-Side Video Experience

When a student logs in with an authorized Gmail and navigates to a class:

- The class appears in “My Classes” with a thumbnail and subject tag.
- Clicking opens an embedded YouTube iframe player inside the portal — no external redirect.
- Player controls: play/pause, seek, fullscreen, playback speed.
- Progress is auto-saved so students can resume where they left off.
- Below the video: class notes download, AI-generated summary, Q&A section, and related resources.
- If access is not granted: a locked card appears with instructor contact info and “Request Access” button.

4. Admin Panel – Full CRUD Operations

Every module in the Aynix AI portal has full Create, Read, Update (Edit), and Delete capabilities for administrators. Below is a comprehensive breakdown of all admin-controllable entities.

4.1 Student Management

Action	Description
Add Student	Register a new student with name, Gmail, roll number, batch, phone, parent contact, profile photo
Edit Student	Update any student detail; re-assign to different batch or program
Delete Student	Soft-delete (deactivate) with option to permanently remove after data export
Bulk Import	Upload CSV to create multiple students at once
View Profile	Full student profile: attendance %, grades, class access list, activity log
Reset Password	Admin can trigger password reset email to student Gmail
Grant/Revoke Class Access	Control which YouTube class videos the student can access

4.2 Faculty / Instructor Management

Action	Description
Add Faculty	Add instructor with name, Gmail, subject specialization, assigned batches
Edit Faculty	Update faculty details, re-assign batches or subjects
Delete Faculty	Deactivate faculty account; reassign their classes
Assign Subjects	Map instructor to subjects/batches
View Performance	See classes delivered, attendance rates, student feedback scores

4.3 Batch & Subject Management

Action	Description
Create Batch	Define batch name, program, start/end date, assigned students and instructor
Edit Batch	Update batch details, add/remove students or instructors

Action	Description
Delete Batch	Archive batch with full history preserved
Create Subject	Add subject with code, name, credit hours, assigned instructor
Edit/Delete Subject	Modify or remove subjects; reassign enrolled students

4.4 Assignments & Assessments

Action	Description
Create Assignment	Title, description, due date, file upload option, max marks, batch/subject
Edit Assignment	Update deadline, marks, description after publishing
Delete Assignment	Remove assignment (student submissions preserved for record)
Grade Submissions	Mark each submission, add feedback, return grades
Create Quiz/Test	MCQ or descriptive questions, time-limited, auto-grading for MCQ
Edit/Delete Quiz	Modify questions, update time, or unpublish quiz
View Results	Per-student and batch-level performance analytics

4.5 Announcements & Notifications

Action	Description
Post Announcement	Push announcement to all students, specific batch, or individual student
Edit Announcement	Update text, attachment, or target audience
Delete Announcement	Remove from all student feeds immediately
Schedule Announcement	Set future publish date/time for timed announcements
Email + In-App	Announcements delivered both via Gmail and in-portal notification bell

5. AI-Powered Features

The Aynix AI portal integrates artificial intelligence meaningfully across the student experience, instructor tools, and administrative functions. All AI features are powered by a Python/Flask microservice that interfaces with OpenAI or Anthropic Claude APIs.

AI Feature	Description	Powered By
AI Class Summarizer	Generates a structured summary of each class from transcript or video title + notes. Students can read instead of rewatch.	Claude / OpenAI
Smart Q&A Chatbot	Per-subject AI chatbot trained on class notes, syllabus, and uploaded PDFs. Students ask questions and get instant answers.	RAG + Claude API
Assignment Auto-Feedback	AI reviews student submissions and provides preliminary feedback on grammar, structure, and content coverage before instructor grading.	OpenAI GPT
Attendance Risk Alert	AI identifies students falling below 75% attendance threshold and auto-generates personalized alert messages for admin to send.	ML / Rule Engine
Personalized Study Plan	Based on attendance, test scores, and weak subjects, AI generates a personalized weekly study schedule per student.	Claude API
Plagiarism Detection	AI compares assignment submissions within the batch to flag potential plagiarism with similarity scores.	Cosine Similarity + NLP
Auto-MCQ Generator	Instructor inputs a topic or uploads notes; AI generates multiple-choice questions for quizzes automatically.	OpenAI GPT-4
Performance Prediction	Predicts likely final grade of each student based on mid-term scores, attendance, and assignment completion trends.	ML Regression Model
Smart Search	Semantic search across all classes, notes, and announcements. Natural language queries like "find class on Python loops".	Embedding + Vector DB
Voice-to-Text Notes	Students can record voice notes during a class; AI transcribes and saves to their notebook.	Whisper API
AI-Generated Reports	Admin can request AI-generated narrative summaries of batch performance, attendance trends, or engagement metrics.	Claude API

AI Architecture Note

AI features run as a separate Python/Flask microservice connected to the main Node.js backend via REST. Class notes and uploaded PDFs are chunked and stored in a vector database (Pinecone or Chroma) for RAG-based Q&A. All AI API calls are rate-limited and cached to manage costs. Admin can enable/disable individual AI features per institute from the settings panel.

6. Additional Essential Modules

6.1 Student Dashboard

- Personalized home with: today's classes, upcoming deadlines, attendance summary widget, recent announcements, AI study tip of the day.
- Quick links: My Classes, Attendance, Assignments, Results, Timetable, Chat.
- Notification bell with real-time alerts via Socket.io.
- Dark/Light mode toggle.

6.2 Timetable / Schedule Module

- Visual weekly timetable per batch/student.
- Admin can create, edit, and delete timetable slots.
- Google Calendar sync — events pushed to student's personal calendar.
- Color-coded by subject; hover/click to see instructor, room, and Meet link.

6.3 Results & Gradebook

- Per-student grade card with subject-wise marks, GPA/CGPA calculation.
- Graphical performance trend (line/bar charts) over semesters.
- Admin can publish/unpublish results per batch.
- Students can download their grade report as PDF.
- Edit/delete marks entries with change-log audit trail.

6.4 Library / Resource Center

- Upload and organize study materials: PDFs, PPTs, links, videos.
- Tag by subject, batch, and topic. Full-text search.
- Admin/instructor can add, edit, delete resources.
- AI-powered: auto-tag new uploads, suggest related resources to students.

6.5 Communication – Internal Messaging

- Student ↔ Instructor direct messaging.
- Instructor can broadcast message to entire batch.
- Admin can message anyone in the system.
- All messages archived, searchable; edit/delete own messages.

6.6 Fee Management

- Admin adds fee structure per batch/program.

- Mark student payments (manual or integrate payment gateway like Razorpay).
- Students see fee dues, receipt download.
- Auto-reminder emails for overdue payments.
- Edit/delete fee records with audit log.

6.7 Settings & Configuration

- Institute profile: name, logo, address, contact.
- Academic year / semester configuration.
- Email template customization for notifications.
- Enable/disable AI features per module.
- API key management (Google, YouTube, AI providers).
- Role permissions matrix editor.

8. 20-Day Project Timeline

The sprint is divided into 4 phases of 5 days each. Work streams run in parallel across frontend, backend, and AI/integration tracks.

Phase 1: Foundation & Core Auth (Days 1–5)

Day	Task	Owner
Day 1	Project setup: repo, CI/CD, DB schema design, Figma design system, component library init	All
Day 1–2	Backend: User auth (JWT + Google OAuth), role-based middleware, base API structure	Emp 2
Day 1–2	Frontend: React scaffold, routing, Tailwind setup, auth pages (Login/Register with Google)	Emp 1
Day 2–3	Database: Students, Faculty, Batches, Subjects, Classes tables; seed data	Emp 2
Day 3–4	Admin Panel: Student CRUD, Faculty CRUD, Batch/Subject CRUD (backend + frontend)	Emp 3 + Emp 5
Day 4–5	Student Dashboard layout, sidebar navigation, notification system (Socket.io)	Emp 1
Day 5	Integration test: auth flow, role access, admin CRUD. Fix blockers.	All

Phase 2: Classes, Video & Attendance (Days 6–10)

Day	Task	Owner
Day 6	YouTube API integration: video metadata fetch, embed player with auth check	Emp 4
Day 6–7	Class module: Create/Edit/Delete class (backend + UI), YouTube URL input, thumbnail	Emp 3 + Emp 1
Day 7	Gmail-based access control: access list per video, grant/revoke UI, email notification via Gmail API	Emp 4
Day 7–8	Student class feed: show only authorized classes per logged-in Gmail; locked state for unauthorized	Emp 1

Day	Task	Owner
Day 8–9	Google Meet integration: create Meet events via Calendar API, attendance pull after session ends	Emp 4
Day 9–10	Online attendance UI: instructor session dashboard, auto-attendance display, manual override, edit/delete	Emp 1 + Emp 3
Day 9–10	Offline QR attendance: QR generation API, rotating token, student scan UI, live instructor dashboard	Emp 4 + Emp 1
Day 10	Attendance reports: cumulative %, export to Excel/PDF, risk alert flags	Emp 3 + Emp 5

Phase 3: AI Features, Assignments & Results (Days 11–15)

Day	Task	Owner
Day 11	Python/Flask AI microservice setup; OpenAI/Claude API integration; vector DB setup for RAG	Emp 4
Day 11–12	AI Class Summarizer: generate summary from notes; display below video in portal	Emp 4 + Emp 1
Day 12	Smart Q&A chatbot: PDF ingestion, chunk embedding, RAG retrieval, chat UI	Emp 4
Day 12–13	Assignment module: Create/Edit/Delete assignment (backend + UI), file submission, grading UI	Emp 3 + Emp 1
Day 13	AI Assignment Feedback: auto-feedback on submission before grading	Emp 4
Day 13–14	Quiz/Test module: MCQ builder, timer, auto-grading, results view	Emp 3
Day 14	AI MCQ Generator: topic input → auto-generate questions	Emp 4
Day 14–15	Results & Gradebook: grade entry, GPA calc, charts, PDF export, edit/delete marks	Emp 2 + Emp 1
Day 15	Timetable module + Google Calendar sync; Library/Resource Center CRUD	Emp 3 + Emp 5

Phase 4: Communication, Polish & Deployment (Days 16–20)

Day	Task	Owner
Day 16	Internal messaging module (student ↔ instructor ↔ admin); broadcast messages to batch	Emp 2 + Emp 1
Day 16–17	Fee management module: fee structure, payment tracking, overdue reminders, receipt download	Emp 3
Day 17	Settings & configuration panel: institute profile, API keys, feature toggles, permission matrix	Emp 2 + Emp 3
Day 17–18	Remaining AI features: attendance risk alert, personalized study plan, performance prediction, smart search	Emp 4
Day 18	UI/UX polish: responsive design (mobile-first), dark/light mode, accessibility, loading states	Emp 5
Day 18–19	End-to-end QA: test all user flows for all roles; bug fixing; cross-browser testing	Emp 5 + All
Day 19	Performance optimization: lazy loading, API caching (Redis), image compression, DB query optimization	Emp 2 + Emp 4
Day 20	Staging deployment, DNS config, SSL, final UAT with client, documentation handover	All

9. Deliverables & Handover

Deliverable	Description	Due Day
Source Code	Full codebase (frontend + backend + AI service) in private GitHub repo	Day 20
Database Schema	SQL schema file + ER diagram	Day 5
API Documentation	Postman collection + Swagger docs for all REST endpoints	Day 20
Deployment Guide	Step-by-step setup guide for local, staging, and production	Day 20
Admin User Manual	PDF guide for admin panel usage with screenshots	Day 20
Student User Guide	PDF guide covering portal features from student perspective	Day 20
AI Service Docs	Architecture, prompt engineering notes, vector DB setup guide	Day 20

10. Risk Register & Mitigations

Risk	Likelihood	Impact	Mitigation
Google Meet API quota limits	Medium	High	Cache attendance data; use service account with higher quota; batch API calls
YouTube embed blocked by institute network	Low	High	Provide fallback: direct portal video upload (Cloudinary) as alternative
AI API cost overrun	Medium	Medium	Rate limiting, response caching, token budget per student/day; use cheaper models for bulk tasks
Scope creep beyond 20 days	Medium	High	Strict change-freeze after Day 10; defer V2 features to backlog
QR code proxy attendance	Low	Medium	One-scan-per-device enforcement + GPS validation + session token expiry
Gmail OAuth permission denial	Low	High	Fallback to username/password auth; Gmail used only for access control, not mandatory

11. Appendix – Complete Feature Checklist

The following is a master checklist of all features to be delivered in the Aynix AI Student Portal v1.0:

Authentication & Security

- Google OAuth 2.0 login (Gmail-based identity)
- JWT token auth with refresh token rotation
- Role-based access control (RBAC) – Admin, Instructor, Student
- Session management with Redis
- Rate limiting & brute-force protection
- HTTPS enforcement + CORS configuration

Admin Panel – All CRUD

- Student: Add, Edit, Delete, Bulk Import, View Profile
- Faculty: Add, Edit, Delete, Assign Subjects
- Batch: Create, Edit, Delete, Assign Students
- Subject: Create, Edit, Delete
- Class/Video: Create, Edit, Delete, Access Control
- Assignment: Create, Edit, Delete, Grade
- Quiz: Create, Edit, Delete, View Results
- Announcement: Post, Edit, Delete, Schedule
- Fee: Create, Edit, Delete, Mark Paid
- Timetable: Add, Edit, Delete slots
- Library: Upload, Edit, Delete resources

Attendance

- Online: Google Meet API integration, auto-pull, duration threshold, manual override, edit, delete
- Offline: QR generation, rotating tokens, student scan, live dashboard, manual entry, edit, delete
- Cumulative reports, export (Excel/PDF), risk alerts

Classes & Video

- YouTube unlisted video integration, portal-embedded player
- Gmail-based per-student access control
- Email notification with portal deep-link
- Watch progress tracking, resume playback
- Notes download, AI summary, Q&A section
- Playlist view by subject/batch

AI Features

- Class summarizer, Smart Q&A chatbot, Assignment feedback
- Attendance risk alert, Study plan generator, Plagiarism detection
- MCQ auto-generator, Performance prediction, Smart search
- Voice-to-text notes, AI-generated narrative reports

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